

Self-Study Assignment 2

Canonical LR(1) / CLR(1) Parser Construction

Course Code: CSE 420

Course Title: Compiler Design

Section: 17

Assignment Topic: Study the construction of the Canonical LR(1), i.e. CLR(1), parser from the reference textbook (Section 4.7.1 to Section 4.7.3) and show the full construction process for the assigned grammar.

Instructions:

1. This is a self-study assignment on **CLR(1) parser construction**.
2. Each student must solve **only one** assigned problem.
3. Let the last two digits of the student's ID be N .
4. Compute:

$$k = N \bmod 20$$

5. The assigned problem will be selected using the following rule:

k value	Assigned Problem
0	Problem 1
1	Problem 2
2	Problem 3
3	Problem 4
4	Problem 5
5	Problem 6
6	Problem 7
7	Problem 8
8	Problem 9
9	Problem 10
10	Problem 11
11	Problem 12
12	Problem 13
13	Problem 14
14	Problem 15
15	Problem 16
16	Problem 17
17	Problem 18
18	Problem 19
19	Problem 20

What to Do

For the assigned grammar, show the complete CLR(1) parser construction process:

- Augment the grammar.
- Write the necessary **FIRST** sets needed for LR(1) lookaheads.
- Construct the initial LR(1) item set I_0 using **CLOSURE**.
- Construct the remaining LR(1) item sets using **GOTO**.
- Draw the CLR(1) automaton / transition diagram.
- Construct the **ACTION** and **GOTO** parsing table.
- State whether the grammar is CLR(1) without conflict.

Problems

Problem 1

Construct the CLR(1) parser for:

$$S \rightarrow aA \quad A \rightarrow bA \mid c$$

Problem 2

Construct the CLR(1) parser for:

$$S \rightarrow aA \quad A \rightarrow b \mid bA$$

Problem 3

Construct the CLR(1) parser for:

$$S \rightarrow AB \quad A \rightarrow a \quad B \rightarrow b$$

Problem 4

Construct the CLR(1) parser for:

$$S \rightarrow AB \quad A \rightarrow aA \mid a \quad B \rightarrow b$$

Problem 5

Construct the CLR(1) parser for:

$$S \rightarrow Aa \quad A \rightarrow bA \mid c$$

Problem 6

Construct the CLR(1) parser for:

$$S \rightarrow aB \quad B \rightarrow cB \mid d$$

Problem 7

Construct the CLR(1) parser for:

$$S \rightarrow CC \quad C \rightarrow cC \mid d$$

Problem 8

Construct the CLR(1) parser for:

$$S \rightarrow aS \mid b$$

Problem 9

Construct the CLR(1) parser for:

$$S \rightarrow Ac \quad A \rightarrow aA \mid b$$

Problem 10

Construct the CLR(1) parser for:

$$S \rightarrow A \quad A \rightarrow aA \mid bA \mid c$$

Problem 11

Construct the CLR(1) parser for:

$$S \rightarrow aSb \mid c$$

Problem 12

Construct the CLR(1) parser for:

$$S \rightarrow xA \quad A \rightarrow yA \mid z$$

Problem 13

Construct the CLR(1) parser for:

$$S \rightarrow aB \mid bA \quad A \rightarrow c \quad B \rightarrow d$$

Problem 14

Construct the CLR(1) parser for:

$$S \rightarrow Aa \mid Bb \quad A \rightarrow c \quad B \rightarrow c$$

Problem 15

Construct the CLR(1) parser for:

$$S \rightarrow AB \quad A \rightarrow a \mid aa \quad B \rightarrow b$$

Problem 16

Construct the CLR(1) parser for:

$$S \rightarrow aA \mid bB \quad A \rightarrow c \quad B \rightarrow d$$

Problem 17

Construct the CLR(1) parser for:

$$S \rightarrow A \quad A \rightarrow aB \quad B \rightarrow bC \quad C \rightarrow c$$

Problem 18

Construct the CLR(1) parser for:

$$S \rightarrow aA \quad A \rightarrow bC \mid d \quad C \rightarrow e$$

Problem 19

Construct the CLR(1) parser for:

$$S \rightarrow aA \mid aB \quad A \rightarrow c \quad B \rightarrow d$$

Problem 20

Construct the CLR(1) parser for:

$$S \rightarrow AB \quad A \rightarrow a \mid b \quad B \rightarrow c \mid d$$

Submission

1. The assignment must be submitted in **handwritten** form.
2. Write all LR(1) items clearly with lookaheads.
3. Show the CLOSURE and GOTO steps in an organized way.
4. Draw the automaton / state transition diagram neatly.
5. Prepare the final ACTION and GOTO table clearly.